

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics Practical

Module - II

Practical – XI

Roll No.:T054	Name:Priti Yadav
Class:TYIT	Batch:2
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11. Optimization Modeling and Simulation Modeling:

a. Formulate an optimization model (e.g., linear programming, integer programming) to solve a real-world business problem and analyze the optimal solution.

Part A: Optimization Modeling

Business Problem: A company produces two products, Product A and Product B. The company wants to maximize its profit using the available machine and labour hours.

Given data:

Product	Profit per Unit	Machine Hours	Labour Hours
A	40	2	1
B	30	1	2

Available Machine Hours = 100

Available Labour Hours = 80

CODE:

```
import numpy as np
from scipy.optimize import linprog
# Profit per unit
profit = [-40, -30]
# Machine and Labour constraints
A = [ [2, 1],
      [1, 2]]
b = [100, 80]
# Solve Linear Programming Problem
result = linprog(
    profit,
    A_ub=A,
    b_ub=b,
    bounds=[(0, None), (0, None)],
    method='highs')
# Optimal units
A_units = result.x[0]
B_units = result.x[1]
# Maximum profit
max_profit = 40 * A_units + 30 * B_units
print("Optimal Number of Product A:", A_units)
print("Optimal Number of Product B:", B_units)
print("Maximum Profit: ₹", max_profit)
# Resource usage
machine_used = 2 * A_units + B_units
labour_used = A_units + 2 * B_units
print("Machine Hours Used:", machine_used)
print("Labour Hours Used:", labour_used)
```

OUTPUT:

```
Optimal Number of Product A: 40.0
Optimal Number of Product B: 20.0
Maximum Profit: ₹ 2200.0
Machine Hours Used: 100.0
Labour Hours Used: 80.0
```

Conclusion :

The optimization model determined that producing 40 units of Product A and 20 units of Product B gives a maximum profit of ₹2,200. The available machine hours (100) and labour hours (80) are fully utilized. Thus, the model helps the company use its resources efficiently and maximize profit.

b. Use simulation modeling to evaluate different business scenarios, such as capacity planning, inventory management, or pricing strategies, and assess their impact on performance metrics.

Part B: Simulation Modeling

Business Problem: A supermarket wants to decide the appropriate inventory level for a product when daily demand is uncertain.

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
# Generate demand for 30 days
np.random.seed(10)
demand = np.random.randint(15, 51, 30)
print("Daily Demand:")
print(demand)
# Inventory levels
inventory_levels = [30, 40, 50]
results = []
for inventory in inventory_levels:
    sold = np.minimum(demand, inventory)
    shortage = np.maximum(demand - inventory, 0)
    inventory_left = np.maximum(inventory - demand, 0)
    print("\nInventory Level:", inventory)
    print("Day | Demand | Sold | Left | Shortage")
    for i in range(30):
        print(i + 1, "|", demand[i], "|",
              sold[i], "|", inventory_left[i],
              "|", shortage[i])
    total_sold = sold.sum()
    total_shortage = shortage.sum()
    print("Total Sold:", total_sold)
    print("Total Shortage:", total_shortage)
    results.append([
        inventory,
        total_sold,
        total_shortage])
# Create DataFrame
df = pd.DataFrame(
    results,
    columns=["Inventory Level", "Total Sales", "Total
Shortage"])
print("\nSimulation Results:")
print(df)
# Create chart
plt.plot(df["Inventory Level"],
         df["Total Sales"],
         marker="o",
         label="Total Sales")
plt.plot(df["Inventory Level"],
         df["Total Shortage"],
         marker="o",
         label="Total Shortage")
plt.xlabel("Inventory Level")
plt.ylabel("Units")
plt.title("Inventory Level vs Sales and Shortage Priti Yadav
T054")
plt.legend()
plt.grid()
plt.show()
```

OUTPUT:

Daily Demand:

[24 30 15 43 40 44 44 23 24 15 31 26 39 48 23 29 28 20 28 40 28 43 37 45
45 40 27 16 46 42]

Inventory Level: 30

Day | Demand | Sold | Left | Shortage

1	24	24	6	0
2	30	30	0	0
3	15	15	15	0
4	43	30	0	13
5	40	30	0	10
6	44	30	0	14
7	44	30	0	14
8	23	23	7	0
9	24	24	6	0
10	15	15	15	0
11	31	30	0	1
12	26	26	4	0
13	39	30	0	9
14	48	30	0	18
15	23	23	7	0
16	29	29	1	0
17	28	28	2	0
18	20	20	10	0
19	28	28	2	0

...

	Inventory Level	Total Sales	Total Shortage
0	30	806	177
1	40	943	40
2	50	983	0

Conclusion:

The simulation showed that increasing the inventory level increases total sales and reduces shortage when demand is higher than available stock. Comparing inventory levels of 30, 40, and 50 units helps the supermarket understand the effect of different stock levels and make a suitable inventory decision based on demand.

